Guangyi Zou

PhD Student in Mathematics
University of California, Irvine

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Research Interests

Random matrix theory, high-dimensional probability, AI for mathematics, theoretical computer science, and mathematical foundations of data science.

Publications

- **BBP Transition for Deformed Inhomogeneous Random Matrices** *2026*
  *Ruohan Geng, Zhantao He, Dang-Zheng Liu, and **Guangyi Zou**; arXiv:2610.24796; preprint*
- **Feasibility Correction in Linear Programs** *2026*
  *Pedro Abdalla, Roman Vershynin, and **Guangyi Zou**; arXiv:2609.36192; preprint*
- **Poisson statistics of one-dimensional random band matrices** *2026*
  *Jiaqi Fan and **Guangyi Zou**; arXiv:2609.23389; preprint*
- **Total Variation Distance between Product Distributions: an Analytic Proxy** *2026*
  *Ariel Avital, Aryeh Kontorovich, Roman Vershynin, and **Guangyi Zou**; arXiv:2609.21049; submitted*
- **On the Subgaussianity of Quantized Linear Maps: An AI-Assisted Note** *2026*
  ***Guangyi Zou** and Roman Vershynin; arXiv:2605.27563; ICML 2026 AI4Math Workshop*
- **Edge Universality for Inhomogeneous Random Matrices II: Markov Chain Comparison and Critical Statistics** *2026*
  *Dang-Zheng Liu and **Guangyi Zou**; arXiv:2604.20215; submitted*
- **Edge Universality for Inhomogeneous Random Matrices** *2025*
  *Dang-Zheng Liu and **Guangyi Zou**; arXiv:2508.17838; submitted*
- **Revisit the Partial Coloring Method: Prefix Spencer and Sampling** *2025*
  *Dongrun Cai, Xue Chen, Wenxuan Shu, Haoyu Wang, and **Guangyi Zou**; arXiv:2408.13756; COCOON 2025*
- **Outliers for Deformed Inhomogeneous Random Matrices** *2024*
  *Ruohan Geng, Dang-Zheng Liu, and **Guangyi Zou**; arXiv:2407.12182; submitted*
- **Edge Statistics for Random Band Matrices** *2023*
  *Dang-Zheng Liu and **Guangyi Zou**; arXiv:2401.00492; submitted*

Education

Ph.D.	University of California, Irvine	Advisor: Prof. Roman Vershynin	Sep 2025 – Present
M.Sc.	University of Science and Technology of China	Advisor: Prof. Dang-Zheng Liu	Sep 2022 – Jun 2024
B.Sc.	University of Science and Technology of China		Sep 2018 – Jun 2022

Research Experience

- **Research Assistant, University of Science and Technology of China** *Mar 2025 – Jul 2025*
  Supervisor: Prof. Dang-Zheng Liu
- **Research Assistant, Tsinghua University** *Sep 2024 – Jan 2025*
  Supervisor: Prof. Fan Yang
- **Academic Visitor, Peking University** *Jan 2022 – May 2022*
  Host: Prof. Xinyi Li

Awards

Euler Outstanding Promise Award	2026
Anhui Province Outstanding Graduate (top 4%)	2024
Graduate Scholarship (2×10,000 RMB)	2022, 2023
One Silver Prize and Three Bronze Prizes, S.-T. Yau College Student Mathematics Contest	2020, 2021
Yang Yuanqing Education Foundation Scholarship (top 1%)	2020

Leadership and Teaching

- **Teaching Assistant, Probability Theory** *Fall 2021; Spring 2022, 2023, 2024*
  University of Science and Technology of China
- **President, Student Algorithm Club** *Jun 2020 – Jun 2021*
  University of Science and Technology of China

Technical Skills

Programming: Python, C++ **Typesetting:** LaTeX

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